Introduction to Cryptography

Lecture : Euler's Totient Function and Euler's Theorem

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Why Euler's Totient Function?

In cryptography, we frequently need to know:

How many integers are relatively prime to a given integer n?

Euler's Totient Function answers this question.

Notation

Euler's Totient Function is denoted by:

$$\boxed{\phi(n)}$$

Definition of Euler's Totient Function

Definition

$\phi(n)$ is the number of positive integers less than n that are relatively prime to n.

That is:

$$\boxed{\phi(n) = |\{a : 1 \leq a < n,\ \gcd(a, n) = 1\}|}$$

Example: Find $\phi(8)$

Positive integers less than 8 are:

$$1, 2, 3, 4, 5, 6, 7$$

Now identify those relatively prime to 8:

$$1,\ 3,\ 5,\ 7$$

because:

$$\gcd(1, 8) = 1$$

$$\gcd(3, 8) = 1$$

$$\gcd(5, 8) = 1$$

$$\gcd(7, 8) = 1$$

Example: Find $\phi(10)$

Positive integers less than 10:

$$1, 2, 3, 4, 5, 6, 7, 8, 9$$

Numbers relatively prime to 10 are:

$$1,\ 3,\ 7,\ 9$$

Therefore:

$$\boxed{\phi(10) = 4}$$

Totient of a Prime Number

Let p be a prime number.
The positive integers less than p are:

$$1, 2, 3, \ldots, p-1$$

Since p is prime, none of these integers shares a factor with p other than 1.
Therefore:

$$\boxed{\phi(p) = p - 1}$$

Examples for Prime Numbers

For:

$$p = 5$$

$$\phi(5) = 5 - 1 = 4$$

For:

$$p = 11$$

$$\phi(11) = 11 - 1 = 10$$

For:

$$p = 17$$

$$\phi(17) = 17 - 1 = 16$$

Some Values of Euler's Totient Function

n	1	2	3	4	5	6	7	8	9
$\phi(n)$	1	1	2	2	4	2	6	4	6

n	10	11	12	13	14	15	16	17	18
$\phi(n)$	4	10	4	12	6	8	8	16	6

Totient of a Product of Two Primes

Suppose:

$$n = pq$$

where p and q are distinct prime numbers.
Then:

$$\boxed{\phi(n) = \phi(p)\phi(q)}$$

Since:

$$\phi(p) = p - 1$$

and:

$$\phi(q) = q - 1$$

we obtain:

Why is $\phi(pq) = (p-1)(q-1)$?

Consider:

$$n = pq$$

There are:

$$pq - 1$$

positive integers less than pq.
Numbers not relatively prime to pq are multiples of:

$$p$$

or:

$$q$$

Multiples of p:

Deriving the Formula

Therefore:

$$\phi(pq) = (pq - 1) - [(q - 1) + (p - 1)]$$

Simplifying:

$$\phi(pq) = pq - 1 - q + 1 - p + 1$$

$$= pq - p - q + 1$$

Factor:

$$\boxed{\phi(pq) = (p - 1)(q - 1)}$$

Numerical: Find $\phi(21)$

Since:

$$21 = 3 \times 7$$

where 3 and 7 are prime:

$$\phi(21) = \phi(3)\phi(7)$$

$$= (3-1)(7-1)$$

$$= 2 \times 6$$

Therefore:

$$\boxed{\phi(21) = 12}$$

Verify $\phi(21) = 12$

The integers less than 21 that are relatively prime to 21 are:

$$1,\ 2,\ 4,\ 5,\ 8,\ 10,$$

$$11,\ 13,\ 16,\ 17,\ 19,\ 20$$

There are exactly:

$$12$$

such integers.
Therefore:

$$\boxed{\phi(21) = 12}$$

More Totient Numericals

Find:

$$\phi(15)$$

Since:

$$15 = 3 \times 5$$

$$\phi(15) = (3-1)(5-1)$$

$$= 2 \times 4$$

$$\boxed{\phi(15) = 8}$$

Similarly:

$$\phi(35) = (5-1)(7-1) = 4 \times 6 = \boxed{24}$$

Useful Extension: Prime Powers

For a prime p:

$$\boxed{\phi(p^k) = p^k - p^{k-1}}$$

Equivalently:

$$\boxed{\phi(p^k) = p^k \left(1 - \frac{1}{p}\right)}$$

Example:

$$\phi(8) = \phi(2^3)$$

$$= 8 - 4$$

$$\boxed{= 4}$$

General Totient Formula

If:

$$n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$$

then:

$$\boxed{\phi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right)}$$

The product is taken over the distinct prime factors of n.

Numerical: Find $\phi(12)$

Prime factorization:

$$12 = 2^2 \times 3$$

Using:

$$\phi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right)$$

we obtain:

$$\begin{aligned}\phi(12) &= 12 \left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right) \\ &= 12 \times \frac{1}{2} \times \frac{2}{3}\end{aligned}$$

$$\boxed{\phi(12) = 4}$$

Numerical: Find $\phi(36)$

Prime factorization:

$$36 = 2^2 \times 3^2$$

Therefore:

$$\phi(36) = 36\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)$$

$$= 36 \times \frac{1}{2} \times \frac{2}{3}$$

$$= 12$$

Thus:

$$\boxed{\phi(36) = 12}$$

Euler's Theorem

Euler's theorem is one of the most important results in number theory for cryptography.

Euler's Theorem

If:

$$\gcd(a, n) = 1$$

then:

$$\boxed{a^{\phi(n)} \equiv 1 \pmod{n}}$$

Important Condition

Euler's theorem requires:

$$\boxed{\gcd(a, n) = 1}$$

That means:

$$a$$

and:

$$n$$

must be relatively prime.

If they are not relatively prime, this form of Euler's theorem cannot be applied directly.

Example: $a = 2,\ n = 11$

Since 11 is prime:

$$\phi(11) = 10$$

Also:

$$\gcd(2, 11) = 1$$

Euler's theorem gives:

$$2^{10} \equiv 1 \pmod{11}$$

Indeed:

$$2^{10} = 1024$$

and:

$$1024 \bmod 11 = 1$$

Example: $a = 3,\ n = 10$

First:

$$\gcd(3, 10) = 1$$

Next:

$$\phi(10) = 4$$

Therefore, Euler's theorem gives:

$$3^4 \equiv 1 \pmod{10}$$

Since:

$$3^4 = 81$$

and:

$$81 \bmod 10 = 1$$

Idea Behind Euler's Theorem

Consider all numbers less than n that are relatively prime to n:

$$R = \{x_1, x_2, \ldots, x_{\phi(n)}\}$$

Now multiply every element by a, modulo n:

$$S = \{ax_1 \bmod n,\ ax_2 \bmod n,\ \ldots,\ ax_{\phi(n)} \bmod n\}$$

If:

$$\gcd(a, n) = 1$$

then S is a rearrangement of R.

Why is S a Permutation of R?

Because:

1. Each x_i is relatively prime to n.
2. a is relatively prime to n.
3. Therefore:

   $$ax_i$$

   is also relatively prime to n.
4. No two distinct values map to the same residue modulo n.

Hence multiplication by a simply rearranges the reduced residue system.

Euler's Theorem: Product Argument

Because S is a permutation of R:

$$\prod_{i=1}^{\phi(n)} (ax_i) \equiv \prod_{i=1}^{\phi(n)} x_i \pmod{n}$$

Thus:

$$a^{\phi(n)} \prod_{i=1}^{\phi(n)} x_i \equiv \prod_{i=1}^{\phi(n)} x_i \pmod{n}$$

Since every x_i is relatively prime to n, cancellation is valid.
Therefore:

$$\boxed{a^{\phi(n)} \equiv 1 \pmod{n}}$$

Using Euler's Theorem for Large Powers

Suppose we need:

$$a^k \bmod n$$

and:

$$\gcd(a, n) = 1$$

Euler's theorem gives:

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

So large exponents can often be reduced using multiples of:

$$\phi(n)$$

This can greatly simplify modular exponentiation.

Numerical: Find 3^{100} mod 10

We know:

$$\phi(10) = 4$$

and:

$$3^4 \equiv 1 \pmod{10}$$

Now:

$$100 = 25 \times 4$$

Therefore:

$$3^{100} = (3^4)^{25}$$

Thus:

$$3^{100} \equiv 1^{25} \pmod{10}$$

Numerical: Find $7^{100} \bmod 15$

First:

$$\gcd(7, 15) = 1$$

Now:

$$\phi(15) = 8$$

Euler's theorem:

$$7^8 \equiv 1 \pmod{15}$$

Write:

$$100 = 12(8) + 4$$

Therefore:

$$7^{100} = (7^8)^{12} 7^4$$

Continue: $7^{100} \bmod 15$

Calculate:

$$7^2 = 49$$

$$49 \bmod 15 = 4$$

Therefore:

$$7^4 \equiv 4^2 \pmod{15}$$

$$= 16 \bmod 15$$

$$= 1$$

Hence:

$$\boxed{7^{100} \bmod 15 = 1}$$

Alternative Form of Euler's Theorem

From:

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

multiplying both sides by a:

$$\boxed{a^{\phi(n)+1} \equiv a \pmod{n}}$$

This form is often useful in modular arithmetic.

Relation to Fermat's Little Theorem

If $n = p$ is prime:

$$\phi(p) = p - 1$$

Euler's theorem becomes:

$$a^{p-1} \equiv 1 \pmod{p}$$

for:

$$\gcd(a, p) = 1$$

This is exactly:

Fermat's Little Theorem

Thus, Fermat's theorem is a special case of Euler's theorem.

Why is Euler's Totient Function Important in Cryptography?

Euler's Totient Function is important in:

- RSA key generation
- Modular inverses
- Euler's theorem
- Modular exponentiation
- Public-key cryptography

For RSA:

$$n = pq$$

and:

$$\boxed{\phi(n) = (p-1)(q-1)}$$

RSA Connection: Small Example

Suppose:

$$p = 5, \qquad q = 11$$

Then:

$$n = pq$$

$$n = 55$$

Now:

$$\phi(55) = (5-1)(11-1)$$

$$= 4 \times 10$$

$$\boxed{\phi(55) = 40}$$

Classroom Activity

Calculate:

1. $\phi(7)$
2. $\phi(9)$
3. $\phi(14)$
4. $\phi(25)$
5. $\phi(35)$
6. $\phi(77)$

Classroom Activity: Answers

$$\phi(7) = 6$$

$$\phi(9) = 6$$

$$\phi(14) = 6$$

$$\phi(25) = 20$$

$$\phi(35) = 24$$

$$\phi(77) = (7-1)(11-1) = \boxed{60}$$

Euler's Theorem: Practice

Use Euler's theorem where applicable.

1. Find:
$$2^{20} \bmod 7$$
2. Find:
$$3^{40} \bmod 10$$
3. Find:
$$5^{100} \bmod 12$$
4. Determine whether Euler's theorem can be directly applied to:
$$6^{20} \bmod 15$$

Practice Solutions

For:

$$2^{20} \bmod 7$$

$$\phi(7) = 6$$

$$20 = 3(6) + 2$$

Therefore:

$$2^{20} \equiv 2^2 \pmod 7$$

$$\boxed{= 4}$$

For:

$$2^{40} \bmod 10$$

Check the Coprimality Condition

Consider:

$$6^{20} \bmod 15$$

First:

$$\gcd(6, 15) = 3$$

Since:

$$\gcd(6, 15) \neq 1$$

the standard form:

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

cannot be directly applied.

Remember

Always check $\gcd(a, n) = 1$ before using Euler's theorem.

Common Student Mistakes

1. Counting n itself while calculating $\phi(n)$.
2. Assuming:
$$\phi(n) = n - 1$$
for every n.
3. Using:
$$\phi(pq) = (p-1)(q-1)$$
without checking that p and q are distinct primes.
4. Applying Euler's theorem without checking:
$$\gcd(a, n) = 1$$
5. Calculating large powers directly instead of reducing the exponent.

Quick Check

Answer briefly:

1. What does $\phi(n)$ represent?
2. What is $\phi(p)$ when p is prime?
3. What is $\phi(pq)$ for distinct primes p, q?
4. Calculate $\phi(21)$.
5. State Euler's theorem.
6. What condition is required before applying Euler's theorem?
7. How is Euler's theorem related to Fermat's Little Theorem?

Lecture Summary

Today we studied:

- Euler's Totient Function
- Relatively prime integers
- $\phi(p) = p - 1$
- $\phi(pq) = (p - 1)(q - 1)$
- General totient formula
- Euler's theorem
- Modular exponentiation using Euler's theorem
- Relationship with Fermat's Little Theorem
- Importance of $\phi(n)$ in RSA

Formula Recap

Prime p:

$$\boxed{\phi(p) = p - 1}$$

Distinct primes p, q:

$$\boxed{\phi(pq) = (p-1)(q-1)}$$

Prime power:

$$\boxed{\phi(p^k) = p^k - p^{k-1}}$$

General form:

$$\boxed{\phi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right)}$$

Euler's Theorem: Formula Recap

If:

$$\gcd(a, n) = 1$$

then:

$$\boxed{a^{\phi(n)} \equiv 1 \pmod{n}}$$

Alternative form:

$$\boxed{a^{\phi(n)+1} \equiv a \pmod{n}}$$

For prime p:

$$\phi(p) = p - 1$$

therefore:

$$\boxed{a^{p-1} \equiv 1 \pmod{p}}$$

Questions & Discussion

Euler's Totient Function tells us

how many invertible residues exist modulo n,

while Euler's theorem gives their powerful exponentiation property.

Your Feedback Matters!

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Lecture 33 – Euler's Totient Function and Euler's Theorem